Ruhr-Universität Bochum

# Your Thesis Title

## A concise description of your research

**Dissertation**

zur Erlangung des Grades eines
Doktors der Naturwissenschaften
an der Fakultät für Physik und Astronomie
der Ruhr-Universität Bochum

von

**Your Name**

Bochum, October 2026

| | |
|---|---|
| Erstgutachter: | Prof. Dr. First Examiner |
| Zweitgutachter: | Prof. Dr. Second Examiner |
| Datum der Disputation: | TBA |

# Abstract

Replace this text with an abstract that states the research problem, approach principal findings, and significance of the thesis.

# Acknowledgements

Thank the people and organisations that supported the work.

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# About this template

> **This is a template, not a thesis**
>
> This website is built from `redeboer/phd-thesis-template`, a reusable Quarto book template for writing and publishing a PhD thesis. Every chapter, title, and author name you see here is placeholder text that a real thesis replaces. It is kept in the repository so that the template stays continuously buildable and so that you can inspect the result before adopting it.

The template provides the parts of a thesis that are the same for everyone: a book structure with front matter, an HTML website, three PDF variants, an ePub edition, bibliography and cross-reference handling, executable code cells, and a continuous-integration setup that publishes the website to GitHub Pages. It is derived from `redeboer/phd-thesis`, with the thesis-specific prose, figures, branding, and scientific dependencies removed.

## What the example chapters show

The chapters in the sidebar are a tour of the authoring features, not a narrative:

- Chapter 1 and Chapter 5 sketch the expected structure of those chapters.
- Chapter 2 contains a labelled equation, Equation (2.1), demonstrating numbered equation cross-references in both HTML and LaTeX.
- Chapter 3 embeds an executable Python cell whose figure and numbers are generated during the build and cached between runs. It also demonstrates folded code, hidden input, and format-dependent output.
- Chapter 4 shows a cross-referenced table, Table 4.1.
- The References page collects citations such as [1; 2], rendered from `docs/references.bib` through a CSL style. The second entry has 26 authors, so it shows how the IEEE style truncates long author lists with "*et al.*".

The same sources produce the downloadable PDF and ePub editions, including paperback and hardcover PDFs with mirrored margins, and an optional Ruhr University Bochum theme.

## Starting your own thesis

1. Click **Use this template** on the repository page and clone the result.

2. Install Pixi and uv, then run `pixi run bootstrap`. The guided configuration asks for your title, author details, institution, and repository, writes them into the Quarto configuration, and removes the template-only tooling.
3. Replace the example chapters and front matter under `docs/`, add your sources to `docs/references.bib`, and run `pixi run doclive` to preview while you write.

The README covers installation, the available build tasks, and publishing in more detail. After bootstrapping, this page becomes the preface of your own thesis.

# 1 Introduction

State the motivation, research questions, scope, and original contributions of the thesis.

## 1.1. Research questions

1. What is the main question addressed by this thesis?
2. Which gap in existing knowledge does it fill?

## 1.2. Contributions

Summarise the contributions in a form that can be revisited in the conclusion.

# 2 Background

Review the concepts and literature needed to understand the original work. Prefer synthesis over a catalogue of papers, and make the relationship to the research questions explicit.

## 2.1. A sample equation

Equations can be labelled and referenced using Quarto syntax:

$$E = mc^2 \tag{2.1}$$

The relation in Equation (2.1) illustrates a numbered cross-reference.

# 3 Methods

Describe the research design, data, apparatus, software, and analysis choices in enough detail for the work to be assessed and reproduced.

## 3.1. Reproducible computations

Executable Python, Julia, or R cells can be placed directly in `.qmd` files. Add the corresponding runtime dependencies to `pyproject.toml`, use a fixed seed for stochastic calculations, and expose the relevant inputs alongside the results.

The following example creates a small synthetic dataset and fits a straight line. Quarto executes the cell while rendering the thesis and caches its output.

The fitted value can also be inserted into the prose as generated Markdown instead of being copied manually:

The fitted slope is **1.81**.

The generated output can be cited like any static asset; for example, Figure 3.1 shows the result of the calculation above. In a real thesis, the same pattern can generate analysis tables and publication figures directly from the source data.

This page deliberately demonstrates several presentation modes. The setup cell shows its input but suppresses output, the generated sentence hides its input with `echo: false`, the plotting cell is folded in HTML with a custom summary, the page-level code tools allow readers to reveal or inspect source, and the PDF build hides all cell input while retaining results.

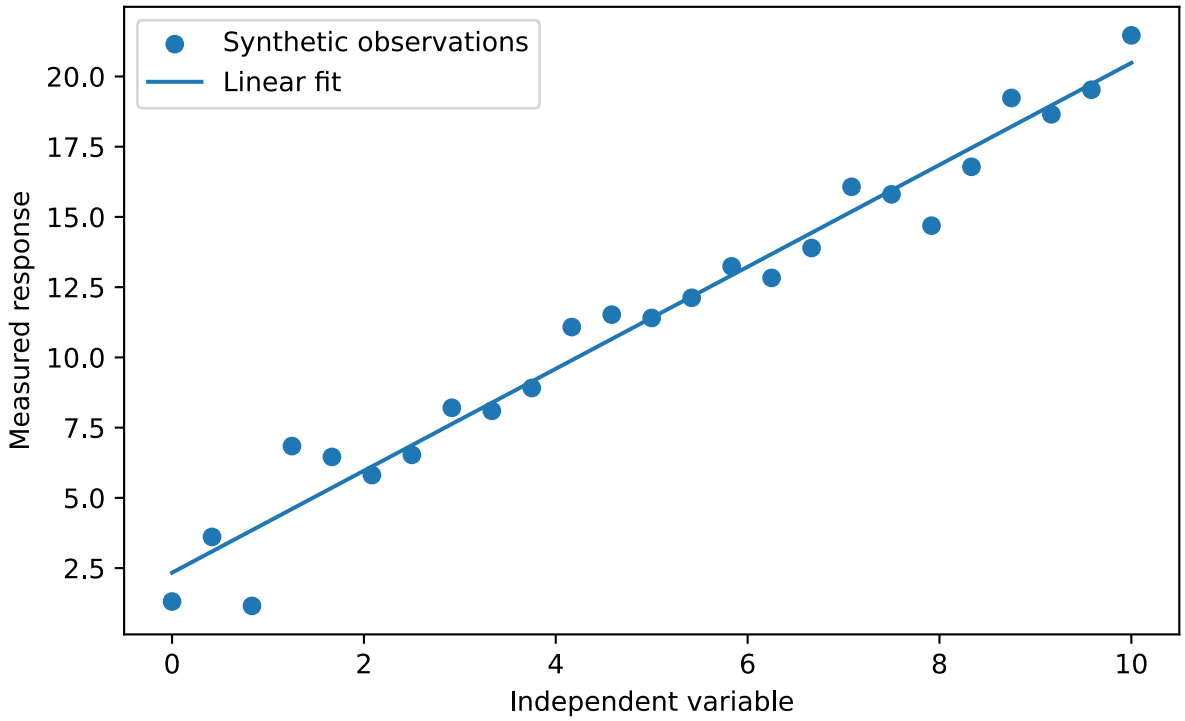



Figure 3.1.: Synthetic observations and the fitted straight-line model. The fixed random seed makes the figure reproducible.

# 4 Results

Present the results in the order needed to answer the research questions. Keep interpretation distinct from measurement where that helps clarity.

| Quantity | Value | Uncertainty |
|---|---|---|
| Example | 1.23 | 0.04 |

Table 4.1.: A sample table.

Refer to it as Table 4.1.

# 5 Conclusion

Answer the research questions, relate the findings to previous work, discuss limitations, and identify concrete directions for future research.

# References


[1] A. Example and B. Researcher, "A placeholder reference for the thesis template," *Journal of Reproducible Examples*, vol. 1, pp. 1–10, 2024.

[2] A. Aalbers *et al.*, "A deliberately overcrowded author list to demonstrate 'et al.' truncation," *Journal of Reproducible Examples*, vol. 2, pp. 11–42, 2025.